Arnab Kumar Roy
Mobile : +91 9851379276
Email : arnab9954@gmail.com

Professional Summary

Software Engineer with 4 years of experience developing modern web applications and AI-powered automation platforms. Specialized in building scalable backend systems with Python, FastAPI, and PostgreSQL, with strong expertise in LLM integration, browser automation, and microservices architecture.

Expert in AI/ML engineering using AWS Bedrock (Claude 3.5 Sonnet), LangChain framework, and RAG architecture with vector databases. Proficient in developing RESTful APIs, implementing Playwright browser automation and building intelligent caching systems with PostgreSQL. Proven track record: 85% reduction in test creation time, 90%+ accuracy in automated processes, and enabling 10+ business domains through scalable solutions. Strong problem-solving skills with ability to work in agile team environments.

Key Skills

Programming Languages & Core Technologies:	Python 3.12, TypeScript, JavaScript, SQL, HTML5, CSS, BDD/Gherkin
Backend Development & Frameworks:	FastAPI (RESTful APIs, Async/Await, Middleware, Request Validation), Django 5.0, SQLAlchemy ORM, Pydantic (Data Validation), Uvicorn (ASGI Server), HTTPX (Async HTTP), Boto3 (AWS SDK), Bootstrap
AI/ML & LLM Technologies:	AWS Bedrock (Claude 3.5 Sonnet), LangChain Framework, RAG Architecture, Prompt Engineering, Vector Databases (ChromaDB), HuggingFace, Sentence Transformers, Embeddings, NLP, Semantic Search, Cosine Similarity, Machine Learning, Neural Networks, Statistical Analysis, Feature Engineering, Model Evaluation, , Computer Vision Concepts, Deep Learning Architectures
Browser Automation & Testing:	Playwright (Async API, Locator Strategies, Iframe Handling), Selenium WebDriver, BDD/Cucumber, Test Script Generation, Cross-browser Testing, Element Identification, Test Automation Frameworks
Databases & Data Management:	PostgreSQL (Schema Design, Query Optimization), ChromaDB (Vector Store), SQLAlchemy ORM, Pandas (Data Processing), Database Migrations, Caching Strategies
APIs, Integration & Protocols:	REST API Design & Development, GitHub API Integration, Jenkins API, Swagger/OpenAPI Documentation, MCP (Model Context Protocol), SSE (Server-Sent Events), JSON Processing, API Authentication, Webhooks
Software Architecture & Design:	Microservices Architecture, Layered Architecture, Async/Await Programming, Error Handling Patterns, Dependency Injection, Design Patterns (Factory, Strategy, Adapter, Repository), Clean Code Principles, SOLID Principles
Development Tools & Version Control:	Git (Branch Management, Merge Strategies), GitHub (Workflows, API Integration, PR Automation), Visual Studio Code, Cursor AI, Docker (Containerization, Multi-stage Builds), Postman (API Testing), Swagger UI, PostgreSQL Admin Tools
CI/CD & DevOps:	Docker Containerization, Automated Testing, Branch Management, Environment Configuration, Continuous Integration
ML Frameworks & Libraries (Currently Learning):	PyTorch, TensorFlow, Keras, PySpark, Scikit-learn, Jupyter Notebooks, DataBricks (Concepts)

Data Science & ML (Currently Learning):	Python Data Science, Pandas (Data Analysis), NumPy, Machine Learning Algorithms, Statistical Analysis, Feature Engineering, Model Training, Data Preprocessing, Exploratory Data Analysis (EDA)

Certifications

Python for Beginners: Learn Python Programming (Python 3) – Udemy Course Completion Certificate | Udemy Business

Professional Experience

Company: COGNIZANT (Dec 2021 – Till now) as Software Engineer

Project Name: TestGenie - AI-Powered Test Automation Platform (December 2024 to Present)

Client: Chewy Inc. (E-commerce Leader - $12B+ Revenue)

Role: AI/ML Software Engineer

Project Description:

Architected and developed TestGenie, an enterprise-grade AI-powered test automation platform that transforms BDD (Behavior-Driven Development) feature files into executable Playwright TypeScript test scripts, reducing test creation time by 85% and enabling rapid QA automation across 10+ business domains.

Key Achievements & Responsibilities:

• Developed AI-powered test generation system using AWS Bedrock (Claude 3.5 Sonnet) with custom LangChain prompts, achieving 90%+ accuracy in automated test script creation and processing 1000+ scenarios monthly

• Built RAG (Retrieval-Augmented Generation) pipeline using ChromaDB vector database with HuggingFace embeddings (sentence-transformers), implementing intelligent caching that reduced context retrieval time by 70%

• Architected RESTful FastAPI microservices with 20+ async endpoints, Swagger documentation, and beautiful error handling, supporting concurrent processing of multiple feature files with real-time progress tracking

• Engineered dual browser automation system with Playwright (JavaScript crawler) and MCP Protocol integration, implementing 15+ intelligent locator generation strategies (getByTestId, getByRole, semantic matching) with iframe handling and cross-platform compatibility

• Designed intelligent locator reuse system with PostgreSQL backend, implementing semantic matching algorithms (80%+ similarity threshold) and multi-attribute validation, reducing script generation time by 60%

• Built end-to-end GitHub/Jenkins CI/CD integration with automated branch versioning, PR creation via REST API, step deduplication against existing repository code, and parameterized Jenkins test execution

• Integrated AWS services including Bedrock for LLM inference with IAM role-based authentication, S3 for file storage operations, and configured AWS SDK (Boto3) for secure service communication

• Developed analytics tracking system storing execution metrics in PostgreSQL, capturing success rates, performance data, and failure analysis for continuous improvement and reporting

• Created email/Slack notification system with HTML-formatted summaries, video recording management with S3 streaming, and team-specific routing for 5+ engineering teams

Technologies: Python, FastAPI, AWS Bedrock (Claude 3.5 Sonnet), LangChain, Playwright, TypeScript, PostgreSQL, ChromaDB, Docker, Jenkins, GitHub API, MCP Protocol, S3, Vector Databases, RAG, Async Programming

Business Impact:

- 85% reduction in test script creation time (hours → minutes)

- 90%+ element identification accuracy across diverse applications

- 60% performance improvement through intelligent caching

- 100% automation of GitHub workflow (branch, PR, Jenkins trigger)

- Enabled non-technical stakeholders to create automated tests

- Accelerated release cycles across 10+ business domains

Project Name: Mars, I-Material EU deployment (February 2023 to December 2024)

Client: Mars

Mars has been a family – owned business for over a century. As a global company with the footprint of a small country, we have the responsibility - and the opportunity - to leave a lasting impact on the world. As a family-owned business, we have the ability to think in generations, rather than just business quarters and have our **Purpose** to guide us on our way. What we do is only as good as how we do it, a goal reflected in our Purpose. We challenge our Associates, partners and suppliers to join us in transforming the way we do business every day.

Project Description:

As a key contributor to the Material EU deployment, I successfully analyzed the Business Requirements Document (BRD), meticulously identifying data discrepancies. My responsibilities extended to creating the Global Framework Table and conducting rigorous testing for the multiple clusters.

During this project, I gained in-depth knowledge of the MDG system and the MM data model. My understanding of the intricate authorization process allowed me to navigate complex scenarios seamlessly. Additionally, I collaborated closely with the lead to craft comprehensive Functional Specification (FS) documentation and also prepared a data comparison script tool using Python which removes 5/6 hours of manual data comparison using excel to just minutes of work that not only saved time and labor but also helped the whole project team to deliver faster within the given time.

Furthermore, I actively participated in the deployment process, swiftly implementing immediate changes to enhance system efficiency.

Project Name: SAW / Stand Alone Website (November 2022 to February 2023)

Project Description:

In this project, I developed a robust e-commerce web application for a single seller. The key features of this application include:

- Custom User Authentication: Implemented a secure and custom user authentication system, ensuring a seamless login and registration process for customers.
- Product Database: Created a comprehensive product database, allowing for efficient management and display of various products.
- Cart & Checkout Functionalities: Designed an intuitive shopping cart system, enabling customers to easily add, review, and modify their selected items. The checkout process is streamlined for a quick and easy transaction.
- Razorpay Payment Integration: Integrated Razorpay, a leading payment solution, to handle customer payments securely and efficiently. This feature ensures a smooth and trustworthy payment experience for customers.

This project demonstrates my ability to create a full-fledged e-commerce application, showcasing my skills in web development, database management, user experience design, and payment system integration. It stands as a testament to my commitment to creating efficient and user-friendly digital solutions.